

Chen-Han Lin

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Education

- National Yang Ming Chiao Tung University**, College of Computer Science Sep 2023 – Jul 2026
M.S. in Data Science and Engineering
- **Coursework:** Machine Learning, Deep Learning, Data Science Project, Biostatistics
- National Cheng Kung University**, College of Science Sep 2018 – Jun 2022
B.S. in Mathematics
- **Coursework:** Probability and Statistics, Fourier Analysis, Data Structures and Algorithms

Master's Thesis

- Model Enhancement and Implementation of Human Pose Estimation Based on the Mamba State Space Architecture**
- **Advisor:** Prof. Chin-Tien Wu
 - Integrated Mamba sequence blocks into a Video Pose Transformer for GPU-accelerated monocular 2D-to-3D pose estimation; reduced inference latency and training time through CUDA optimization, parallel data pipelines, and modular integration.

Internship Experience

- Whale Force Holdings Ltd.**, Data Scientist Intern Mar 2026 – Jul 2026
- Systematic Strategy Research:**
- Built a modular strategy evaluation framework establishing a statistical validation protocol: each rule's excess returns and Sharpe ratio tested against **10,000-path** Monte Carlo random-pick null distributions to separate significant alpha from selection luck.
- Financial Statement Parsing:**
- Built a financial-statement parsing pipeline via rule-based XBRL tree traversal, recovering line items **missed by the official SEC company-facts API** and validating extraction against OCR ground truth.
- Institutional Portfolio Research:**
- Clustered institutional investors from **44 quarters** of raw SEC 13F filings via engineered portfolio features (concentration, breadth, turnover) and direct holdings-weight vectorization.
 - Validated cluster quality across silhouette, Calinski-Harabasz, and Davies-Bouldin indices, yielding interpretable investor-style segments.

Projects

- LoRA and Combined Parameter-Efficient Tuning for Large Models** [🔗](#) Jun 2024
- Investigated the integration of parameter-efficient fine-tuning (PEFT) methods such as LoRA, BitFit, Adapter, and Prefix-tuning across both vision and NLP tasks.
 - Demonstrated that **LoRA + Adapter** achieved a **+3.6% improvement in accuracy** on vision classification tasks (from 0.8775 to 0.9134), with only **1.6% trainable parameters**.
 - Showed that combining LoRA and BitFit improved NLP accuracy on RoBERTa-base from 0.931 to **0.945** (SST-2), confirming the complementary nature of these PEFT methods.
- Data Science Project: Pima Indian Diabetes Dataset** [🔗](#) Mar 2025 - Jun 2025
- Owned the full pipeline from raw data to conclusions: built classification pipelines across logistic regression, decision trees, random forests, and XGBoost, combining preprocessing, domain-specific feature engineering, stratified cross-validation, and grid search.
 - Evaluated models with ROC-AUC, precision, recall, F1-score, and calibration curves, achieving **Accuracy = 83.6%, ROC-AUC = 91.2%, Precision = 84.5%, Recall = 88.2%, and F1 = 86.3%**.

Selected Honors & Competitions

TSMC IT CareerHack

Feb 2026

1st Place ranking

- Built a Docker-isolated refactoring platform with LangChain Deep Agents, FastAPI, React, and MongoDB, automating iterative analysis, refactoring, testing, and quality evaluation across three legacy repositories.

WorldQuant International Quant Championship

Mar 2026 - Jun 2026

4th Place in Taiwan

- Built a closed-loop alpha research engine that encoded backtest results as graph constraints, pruned redundant hypotheses, and re-ranked signals by marginal portfolio contribution.
- Evaluated **5,164** alpha expressions, with **1,023** passing all seven platform checks, and helped construct a **52-alpha** portfolio that finished **4th in Taiwan**.

TSMC IT CareerHack

Jan 2024

4th Place ranking

- Fine-tuned LLaVA with LoRA for task-specific visual question answering, supported by targeted data augmentation and noise filtering.

TSMC Intelligent Manufacturing Workshop

Mar 2025

Top-Performing Team, Scheduling Optimization

- Formulated the problem as a graph-based job-shop optimization, converting locations to nodes and distances to weighted edges, and applied Integer Linear Programming (ILP) to obtain the optimal schedule.

AI CUP 2024 Image Data Generation for UAVs

May 2024 - Jun 2024

Honorable Award

- Built a mask-conditioned image-generation pipeline with targeted augmentation and hyperparameter tuning, evaluated using Fréchet Inception Distance (FID).

AI GO Skyrocketed Stocks Forecasting

Mar 2025 – Apr 2025

- Built and tuned XGBoost, Random Forest, and TabNet models for stock-movement prediction using feature selection and tabular feature interactions.

Technical Skills

Programming, Data & Machine Learning:

- Python, C, C++, SQL, Bash; NumPy, pandas, scikit-learn, PyTorch, Transformers, TabNet, and Mamba
- Data preprocessing, feature engineering, ETL pipelines, and reproducible experimentation

Quantitative Research:

- Alpha research, backtesting, Monte Carlo simulation, portfolio and benchmark analysis, risk-adjusted performance evaluation, and statistical modeling
- Mathematical optimization using Integer Linear Programming and graph-based optimization methods

AI Engineering & Agentic Workflows:

- Coding agents, agent skills and harnesses, tool use, repository-level code generation, refactoring, debugging, and automated test generation
- Context engineering, task decomposition, workflow orchestration, iterative execution, agent evaluation, and human-in-the-loop validation

Software Engineering & Infrastructure:

- Linux, Git, Docker, Google Cloud Platform (GCP), Amazon Web Services (AWS), and MATLAB
- Modular system design, version control, debugging, performance profiling, and Linux-based data pipelines